

Santander publishes AI projects under an Open Source licence to ramp up shared innovation



Santander is opening up its technology to help build more trustworthy, competitive artificial intelligence (AI) in banking, on a par with sector leaders. The bank has shared over a dozen of its AI projects under an Open Source licence, enabling developers, researchers and professionals the world over to use this technology free of charge to grow. You can check out what Santander has published on its new channel on GitHub, the global repository of shared initiatives.

Banco Santander remains at the forefront of AI development by offering public repositories on Open Source to foster technical cooperation and shared learning. The 11 projects it has chosen apply directly to banks and other businesses and could bring benefits to several sectors. The aim is to work with the community on developing the AI of the future.

From now on, other developers will be able to put forward proposals to enhance Santander AI Lab's tools, thus optimizing a model of lifelong learning and development based on shared intelligence. According to José Manuel de la Chica, Head of Santander AI Lab, "we're not doing this because Open Source is the talk of the town. We're doing it because the true challenges of advanced AI — security, equity, robustness, privacy, governance and traceability — are too important and cross-cutting to address them in silos".

At Santander, we believe that the next phase of artificial intelligence will not just depend on who has access to the most advanced models, but on who is capable of using them with rigour, confidence and responsibility. In banking, this means being able to prove that systems are secure, fair, robust and auditable. That's why we decided to open up some of the Santander AI Lab's work — special tools, without real

customer data, which can help the community make headway with some of AI's toughest challenges. We want to contribute, learn and cooperate from a very clear position: technological ambition, acting responsibly at all times, and with our feet firmly on the ground.

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José Manuel de la Chica, Head of Santander AI Lab

This initiative enables us to share tools, examples and resources developed by the bank's teams in such areas as AI, machine learning, large language models, generative AI, responsible AI, and AI governance. Publishing these projects is part of Santander's commitment to responsible innovation, cooperation with the technology ecosystem, and exchange of technical knowledge under internal intellectual property, data protection, cybersecurity, licensing and brand review processes. Each repository includes technical documents, an Apache 2.0 Open Source licence, contribution guides, codes of conduct, security information, and review processes.

Synthetic data for fraud detection

Santander AI Lab has published the gen-fraud-graph on GitHub. It's a tool that helps create synthetic networks of fraud-related transactions and behaviours. It addresses one of banking's major AI challenges: learning to uncover complex patterns without compromising people's privacy. The idea is simple: if we want to enhance fraud detection, we need environments where we can test hypotheses, compare models, and understand how certain signs emerge. This project generates artificial data to mirror specific risk patterns. There are no real customer data as “AI innovation must run alongside privacy, not at its expense”.

Transparent rules

We have also shared mech-gov-framework, which explores how to build governance mechanisms based on language models, especially when they can influence sensitive or high-impact decisions. It's a key initiative as “it attempts to enable the rigorous review, traceability and governance of AI functioning”. The aim is to turn security, consistency, auditability, control and other principles into more specific and measurable elements. Rather than blindly trust a system's response, the project proposes working with rules, thresholds and checks that help decide when to go live, under what conditions, and with which limits.

Fair systems

Another project that Santander AI Lab has opened up is mutatis-mutandis, which addresses the challenge of getting AI systems to act fairly and treat people in comparable situations equally. This initiative looks into algorithmic discrimination to enable analysis with counterfactual comparators. The main idea is to move beyond general statements on fairness to tools that enable us to examine, test and discuss the behaviour of models with greater precision, as “trust in AI is built not only by looking at its results, but by understanding how it gets there and whom it may affect”.

Santander AI Lab initiatives on Open Source



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